**2** A particle enters the region between two horizontal metal plates in a vacuum, as shown in Fig. 2.1. While the particle is between the plates, it experiences a constant upward acceleration of 24 m $s^{-2}$.

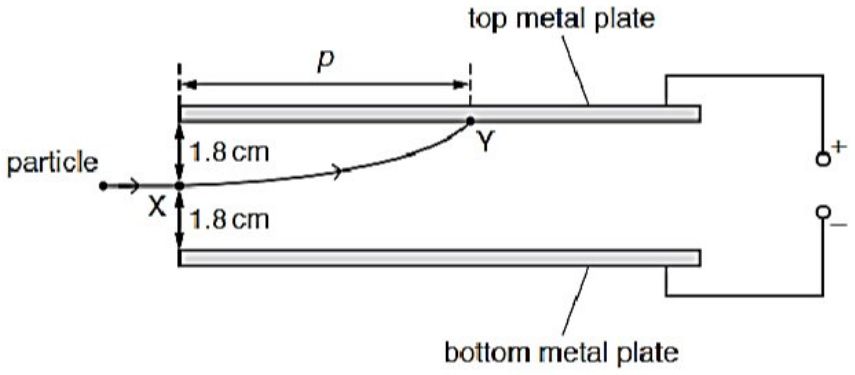



**Fig. 2.1**

The particle enters the electric field at point X with a horizontal speed of 0.75 m $s^{-1}$ along a line midway between the plates.

The separation of the plates is 3.6 cm.

**(a)** Determine the time *t* for the particle to strike the upper plate.

*t* = ..... s [2]

**(b)** Hence determine the minimum length *p* of the plates for the particle to just strike the upper plate.

*p* = ..... cm [1]

**(c)** The plates have length 4.0 cm. The voltage is adjusted such that the net force acting on the particle remains unchanged. Determine the minimum separation between the plates such that the particle does not strike either plate.

minimum separation = ..... cm [3]